

Proposal for One-Day Workshop on Flask Framework

Title: Flask Framework Essentials – A Hands-on Workshop

Organized by: Vidyavardhaka College of Engineering, ISE – B (6th semester)

Target Audience: BE Students

Duration: 5 Hours

Workshop Objective: This workshop aims to introduce students to the Flask web framework and help them develop a foundational understanding of how to build simple web applications. The session will focus on Flask's core features without CRUD operations and authentication, ensuring participants grasp essential concepts through a hands-on approach.

Workshop Schedule & Content

Session 1: Introduction (10 mins)

- Overview of Flask
- Importance and applications of Flask in modern web development
- Examples of real-world Flask applications

Session 2: Application Basics (30 mins)

- Setting up a Flask environment
- Writing a simple "Hello World" web app
- Understanding Flask components (routes, views, templates, and rendering responses)
- Brief explanation of Jinja templating (without authentication features)

Session 3: Database Integration (1 hour)

- Understanding Flask-SQLAlchemy for database management
- Creating a simple **Sailors Database**
- Setting up database models and relationships
- Performing basic queries using Flask (excluding CRUD operations)

Session 4: FAQs & Hands-on Queries (2 hours)

- Addressing participant questions and debugging issues
- Hands-on exercises to reinforce learning

- Demonstration of Flask's capabilities using real-time examples

Session 5: Conclusion (10 mins)

- Summary of topics covered
- Additional resources helpful for their DBMS - ABA
- Feedback collection & closing remarks

Application Example: Sailors Reservation System

- A minimal web application where sailors can view available boats and their details.
- The app will showcase Flask routing, database connectivity, and rendering HTML pages dynamically.

Outcome: Participants will gain a clear understanding of how to develop and structure Flask applications.

Future work scope: Add authentication and additional security features using CRUD operations.

Requirements

- **Software:** Python, Flask, MySQL, VS Code
- **Pre-requisites for Students:** Basic understanding of Python
- **Resources Provided:** Sample code, reference materials, and documentation

Approval Request

We request approval from the respected authorities to conduct this workshop. Kindly review the proposal and grant permission to proceed with the organization of this event.

For further inquiries, please contact:

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